

CEO Voice Agent - Engineering Task

Overview

Build a voice agent that generates social media content (X and LinkedIn) in a specific CEO's authentic voice. The system should take a topic or idea as input and produce platform-specific drafts that sound like the CEO wrote them - close enough that a human editor's work is limited to structural and strategic adjustments rather than voice correction.

This V1 is scoped to two individuals at Databricks: Ali Ghodsi (CEO) and Matei Zaharia (CTO).
Note: we're looking to build a voice agent for all leaders not just CEOs but to keep this task tight - we're specifically looking at CEOs and CTOs and training for them.

Problem Statement

Current LLM-based voice replication follows a predictable pattern: feed past posts into a model, extract a short voice summary, use it as a system prompt. The result is generic. A 10-line voice description captures surface-level traits ("uses short sentences, informal tone") but misses the micro-patterns that make someone's writing recognizable - their sentence length distribution, how they open posts, their rhetorical habits, their platform-specific formatting, their tonal register.

The goal is to build a system that captures voice at a much deeper level and applies it reliably when generating new content.

Architecture

Our recommendation is a retrieval-augmented generation (RAG) approach rather than model fine-tuning for V1, as it is cheaper per client, works with less data, and is easier to update as new content becomes available. That said, the technical approach is ultimately up to you. If you believe fine-tuning, a hybrid approach, or something else entirely will produce better results, you have full latitude to pursue it. What matters is the quality of the output.

The system should have three core components, however you choose to implement them:

1. Voice Profile Engine

For each CEO, build a structured, multi-dimensional voice profile derived from their public content. This is not a paragraph summary. It should capture at minimum:

- Lexical patterns: vocabulary range, recurring phrases, word-level preferences (e.g., "folks" vs. "team" vs. "people"), jargon usage, how technical they go

- Structural habits: sentence length distribution, paragraph length, how they open and close posts, use of lists/line breaks/questions, post length tendencies
- Rhetorical moves: do they use analogies? Do they name-drop? Personal anecdotes vs. data-driven arguments? How do they handle attribution and credit? Do they ask the audience questions?
- Platform-specific formatting: X voice vs. LinkedIn voice (these are almost always different for the same person). Thread structure, emoji usage, hashtag behavior, use of links/media
- Tonal register: warmth vs. distance, humor vs. gravitas, level of personal disclosure, how they express conviction, how they handle vulnerability or uncertainty

The voice profile should be stored in a structured format that the draft generator can reference at generation time. It should be updatable as new content becomes available.

2. Virality Structure Library

A separate module that captures structural and formatting patterns from high-performing content across the top 100 most influential profiles on X and LinkedIn. (This list will be provided separately.)

The purpose of this module is to learn what makes CEO content perform well structurally - hook patterns, post formats, pacing, how stories are told, what call-to-action styles drive engagement. This is platform knowledge, not voice knowledge.

This module informs the draft generator about structure and format, while the voice profile engine informs it about how the specific CEO sounds. These two layers must stay separable. The virality layer should enhance the CEO's natural voice with better structural choices. The degree of influence from this layer should be subtle - think of it as turning the dial 10-15% toward higher-performing structural patterns while keeping the voice fully intact. The output should never feel like it is trying too hard.

Directions to explore (not exhaustive - you have latitude here):

- Cluster high-performing posts by structural pattern (story arc, hot take, thread breakdown, personal reflection, etc.) and learn which patterns correlate with engagement for different CEO archetypes
- Identify hook patterns that consistently outperform across the corpus
- Study how line breaks, spacing, and formatting differ between average and high-performing posts
- Analyze what makes a post feel sharp or compelling without crossing into clickbait or inauthenticity

3. Draft Generator

Takes three inputs:

- Idea/angle: a description of what the post is about and the narrative angle (see Input/Output Examples at the bottom of this document)
- CEO identity: which voice profile to use
- Platform: X or LinkedIn

Produces a draft post (or thread, for X) that combines the CEO's voice with structurally sound formatting.

Data Pipeline

The engineering team is responsible for sourcing all data from publicly available channels. No data will be provided upfront. For each profile (Ali Ghodsi and Matei Zaharia), source content from:

1. X (Twitter) post history - all publicly available posts. This is the primary signal for X voice.
2. LinkedIn post history - all publicly available posts. This is the primary signal for LinkedIn voice.
3. Earnings call transcripts - publicly available through SEC filings, investor relations pages, and financial data providers. These capture how the CEO speaks in a formal, high-stakes context and reveal vocabulary, framing, and how they talk about strategy.
4. YouTube appearances - interviews, keynotes, fireside chats. Transcribe these. They capture the CEO's conversational voice, which often bleeds into their social media writing.
5. Podcast appearances - source transcripts or transcribe audio. Conversational voice, storytelling patterns, how they explain complex ideas simply.
6. Conference and speaking event appearances - keynotes, panels, product launches. Transcribe if needed. These capture how the CEO presents ideas to a live audience, which often maps closely to how they write LinkedIn posts.
7. Published interviews - print and digital media (Forbes, Bloomberg, TechCrunch, etc.). These show how the CEO's words are represented when filtered through a journalist, useful for understanding their public framing.
8. Blog posts and authored content - company blog, personal blog, authored op-eds. If the CEO has a book, include relevant excerpts.
9. Annual shareholder letters - often written personally by the CEO and reveal their long-form voice, values framing, and how they narrate a company's trajectory.

Priority weighting: X and LinkedIn post history should carry the most weight in voice extraction, as they are the exact medium the system is generating for. Spoken content (earnings calls, podcasts, YouTube, events) should be treated as supplementary signal that enriches understanding of how the CEO thinks and talks, which informs but does not dominate the written voice profile.

Editing and Iteration Workflow

The system must support an iterative human-in-the-loop workflow:

1. Generate: User provides an idea and CEO identity. System produces a draft.
2. Edit: Human editor makes structural changes to the draft - rewrites sections, changes the angle, reorganizes the post, adds or removes points.
3. Re-voice: The edited draft is fed back into the voice agent. The system applies the CEO's voice to the new structure and outputs an updated draft. The voice should be as good or better than the first pass, because it now has the human's structural intent as additional signal.
4. Repeat: This loop can run multiple times. Each pass should preserve the human's structural choices while tightening voice fidelity.

The key constraint: the system must distinguish between structural changes the human made intentionally and voice elements. It should not override the human's structural decisions. It should only refine voice.

Evaluation

A. LLM-as-Judge (Automated, for development iteration)

Set up a separate LLM evaluation pipeline. Provide it with 20+ real posts from the CEO/CTO as reference material. For each generated post, have the judge LLM score voice fidelity on a 1-10 scale with written reasoning. This gives the team a fast feedback loop during development and should be running continuously as you build.

B. Manual Review (Primary quality gate)

- 5 generated X posts and 5 generated LinkedIn posts across different topic types (product launch, acquisition, earnings, personal reflection, industry commentary)
- For each post, include the input idea/angle that was provided to the system

I will score each post on three dimensions, each on a 1-5 scale:

- Voice accuracy: does this sound like them and is it slightly better engineered for reach than their previous posting voice?
- Post quality: is this a well-written, well-structured post for the platform (X or LinkedIn)?
- Naturalness: does this feel like something a real person wrote?

Target: average score of 4+ across all three dimensions before V1 is considered complete.

Deliverables

1. Data pipeline that can ingest and process all source types listed above for a given CEO
2. Voice profile engine that produces a structured, multi-dimensional voice profile from the ingested data
3. Virality structure library built from the top 100 influential profiles (list provided separately)
4. Draft generator that takes idea + CEO + platform and produces a draft
5. Re-voicing endpoint that takes a human-edited draft and applies the CEO's voice to it
6. Evaluation results for Ali Ghodsi, Matei Zaharia, and one additional CEO, using the framework above, with scores and sample outputs
7. Technical documentation covering architecture decisions, what worked, what didn't, and recommendations for scaling to additional CEOs
8. A minimal front-end UI is welcome if feasible - something simple where a user can select a CEO, type in an idea, choose a platform, and see the generated draft with an option to edit and re-voice. This is optional but encouraged.

Constraints

- All source data must be publicly available. Do not use any data that requires authentication, payment, or terms-of-service violations to access.
- The system should be designed so that onboarding a new CEO is a matter of running the data pipeline and building a voice profile, not re-engineering the system.
- Platform outputs: X posts (single posts and threads up to 5 posts) and LinkedIn posts.
- The virality layer should be a dial, not a switch. The default should be subtle.

Scope Boundaries

In scope:

- Voice profiling and draft generation for Ali Ghodsi and Matei Zaharia
- X and LinkedIn as output platforms
- Iterative editing workflow
- Evaluation framework and results
- Minimal front-end UI (optional)

Out of scope for V1:

- Real-time posting or scheduling
- Analytics or performance tracking of published posts
- Multi-language support
- Image or media generation

Input/Output Examples

These examples illustrate what we mean by "idea" as an input and what the expected output structure looks like.

Example 1: Acquisition

Input:

- CEO: Ali Ghodsi
- Platform: LinkedIn
- Idea/Angle: "Databricks just acquired Tabular, the company behind Apache Iceberg. The angle is that this validates the open-source approach to data infrastructure. The best technology wins when it is open, and this acquisition brings together the teams behind Spark and Iceberg under one roof."

Expected output: A single LinkedIn post, 150-300 words, written in Ali Ghodsi's LinkedIn voice. Opens with a hook that reflects how he typically starts posts. Weaves in the strategic narrative (open-source winning) with a personal or company-specific detail. Closes in a way that matches his usual LinkedIn closing style (does he ask a question? Make a forward-looking statement? Tag people?). Formatting should match his typical LinkedIn patterns (paragraph breaks, line spacing, emoji usage or lack thereof).

Example 2: Industry Commentary

Input:

- CTO: Matei Zaharia
- Platform: X
- Idea/Angle: "The AI industry is converging on compound AI systems rather than monolithic models. The angle is that the next wave of AI progress will come from how you orchestrate multiple models, retrieval, and tools together, not from making a single model bigger. This is what Databricks has been building toward with Mosaic and their ML platform."

Expected output: A single X post or short thread (2-3 posts), written in Matei Zaharia's X voice. Should reflect his typical level of technicality on X (does he go deep or keep it accessible?). Formatting should match his X conventions (thread structure, use of links, tagging, etc.). The post should feel like a natural thought he would share, not a corporate announcement.

Example 3: The editing loop in practice

1. You give the system Example 1 above. It generates a LinkedIn post for Ali Ghodsi.
2. You read the draft and decide the hook is wrong - you want it to start with a personal story about meeting the Tabular founders, not a statement about open-source. You rewrite the first two sentences and rearrange the middle paragraph.
3. You feed the edited version back into the system with a note: "Re-voice this. Keep my structural changes, refine for Ali's voice."

4. The system outputs an updated version that preserves your new hook and paragraph order but tightens the language to match Ali Ghodsi's voice patterns.

This loop is the core workflow. The system handles voice. The human handles ideas and structure.

A helpful list of impactful X tech CEOs and CTOs (feel free to change and build your own list for both X and LinkedIn):

Sam Altman | @sama | CEO
Brian Chesky | @bchesky | CEO
Soumith Chintala | @soumithchintala | CTO
Garry Tan | @garrytan | CEO
Suhail | @Suhail | CEO
Aaron Levie | @levie | CEO
Guillermo Rauch | @rauchg | CEO
Michael Truell | @mntruell | CEO
hardmaru | @hardmaru | CEO
Amjad Masad | @amasad | CEO
Jason Lemkin | @jasonlk | CEO
Dario Amodei | @DarioAmodei | CEO
Theo | @theo | CEO
Ilya Sutskever | @ilyasut | CEO
Ryan Petersen | @typesfast | CEO
Patrick Collison | @patrickc | CEO
Travis Kalanick | @travisk | CEO
Austen Allred | @Austen | CEO
Dylan Field | @zoink | CEO
Will Manidis | @WillManidis | CEO
Tanishq Abraham | @iScienceLuvr | CEO
Tobi Lutke | @tobi | CEO
David Friedberg | @friedberg | CEO
Clement Delangue | @ClementDelangue | CEO
Patrick OShaughnessy | @patrick_oshag | CEO
Jason Ginsberg | @JasonBud | CTO
Karri Saarinen | @karrisaarinen | CEO
Rahul | @0interestrates | CEO
Andrew Milich | @milichab | CEO
Eric Schmidt | @ericschmidt | CEO
Jordan Singer | @jsngr | CEO
Elon Musk | @elonmusk | CEO
Mira Murati | @miramurati | CEO/CTO
Demis Hassabis | @demishassabis | CEO

Andrew Feldman | @andrewdfeldman | CEO
Josh Miller | @joshm | CEO
Lulu Cheng Meservey | @lulumeservey | CEO
Sundar Pichai | @sundarpichai | CEO
Matt Shumer | @mattshumer_ | CEO
Ivan Zhao | @ivanhzhao | CEO
Karol Hausman | @hausman_k | CEO
Fidji Simo | @fidjissimo | CEO
Carina Hong | @CarinaLHong | CEO
Yuchen Jin | @Yuchenj_UW | CTO
Bryan Johnson | @bryan_johnson | CEO
Daniel Loeb | @DanielSLoeb1 | CEO
Julien Chaumond | @julien_c | CTO
Varun Mohan | @_mohansolo | CEO
Tony Zhao | @tonyzzhao | CEO
Kiana Ehsani | @ehsanik | CEO
Ryan Carson | @ryancarson | CEO
Brad Gerstner | @altcap | CEO
Bill Ackman | @BillAckman | CEO
Steve Ruiz | @steveruizok | CEO
David Singleton | @dps | CEO/CTO
Michael Dell | @MichaelDell | CEO
Asha | @asha_shar | CEO
Erik Bernhardsson | @bernhardsson | CEO
Joon Sung Park | @joon_s_pk | CEO
Richard Socher | @RichardSocher | CEO
Alex Finn | @AlexFinn | CEO
Dustin Moskovitz | @moskov | CEO
Alec Stapp | @AlecStapp | CEO
Keller Clifton | @Keller | CEO
David Ulevitch | @davidu | CEO
Barret Zoph | @barret_zoph | CTO
Dan Shipper | @danshipper | CEO
Immad Akhund | @immad | CEO
Jack Dorsey | @jack | CEO
Alex Cohen | @anothercohen | CEO
Steven Bartlett | @StevenBartlett | CEO
Dharmesh Shah | @dharmesh | CTO
Red Xiao | @Red_Xiao_ | CEO
Mike Krieger | @mikeyk | CTO
Philip Johnston | @PhilipJohnston | CEO
Qasar Younis | @qasar | CEO
Steve (Builder.io) | @Steve8708 | CEO
Rahul | @rahulgs | CTO

Emmett Shear | @eshear | CEO
Bret Taylor | @btaylor | CEO
Max Hodak | @maxhodak_ | CEO
Satya Nadella | @satyanadella | CEO
DHH | @dhh | CTO
Lucy Guo | @lucy_guo | CEO
Brendan Foody | @BrendanFoody | CEO
Tanay Kothari | @tankots | CEO
John Carmack | @ID_AA_Carmack | CTO
Michael Andregg | @michaelandregg | CEO
Alex LeBrun | @lxbrun | CEO
Max Spero | @max_spero_ | CEO
Danielle Fong | @DanielleFong | CEO
Brian Halligan | @bhalligan | CEO
Alap Shah | @alapshah1 | CEO
Cathie Wood | @CathieDWood | CEO
Jason Warner | @jasoncwarner | CTO
Reiner Pope | @reinerpope | CEO
Christoph Nakazawa | @cnakazawa | CEO
Zachary Perret | @zachperret | CEO
Ivan Burazin | @ivanburazin | CEO
Aravind Srinivas | @AravSrinivas | CEO